

# Compression Physics: A Domain-Normalized Efficiency Index (Generalized Efficiency Coefficient)

Mycal Brooks

project: Tower

Email: MycalBrooks@projecttower.org

## Abstract

We introduce the Generalized Efficiency Coefficient (GEC), a dimensionless index measuring distance-to-frontier efficiency across domains, from thermodynamics to machine learning. Defined as  $\text{GEC}_0 = \eta/C_{\max}$ , with  $\eta = Y/X$  (yield over input) and  $C_{\max}$  as the domain’s theoretical, physical, or empirical frontier, GEC yields a normalized  $[0, 1]$  score. A Compression Scaling Kernel (CSK) vector  $\vec{\kappa} = (S, H, D, R, E)$  provides diagnostic decomposition. We contribute: (i) a tiered  $C_{\max}$  catalog with uncertainty quantification; (ii) a CSK with auditable proxies for structural and operational factors; (iii) a Comparability Protocol for rigorous, reproducible benchmarking; (iv) theorems on unit invariance, normalization, and frontier consistency; and optional *verifiable provenance* for empirical bounds. Case studies across five domains demonstrate uncertainty-aware reporting,  $[0, 1]$  normalization, and actionable CSK diagnostics.

**Keywords**— Efficiency, benchmarking, data envelopment analysis, Shannon capacity, exergy, machine learning efficiency.

## 1 Introduction

Efficiency is the engine of progress, yet its metrics are fragmented: Carnot’s limit  $(1 - T_C/T_H)$  governs heat engines [1], Shannon’s capacity  $(B \log_2(1 + \text{SNR}))$  bounds communications [2], and data envelopment analysis (DEA) benchmarks firms [3, 4]. Compression Physics models processes as transformations of inputs into validated outputs. The Generalized Efficiency Coefficient (GEC) normalizes yield ( $Y$ ) per input ( $X$ ) against a domain frontier ( $C_{\max}$ ), with a factor decomposition via a Compression Scaling Kernel (CSK) vector  $\vec{\kappa} = (S, H, D, R, E)$ .

Our contributions are:

- A tiered  $C_{\max}$  catalog blending theoretical, physical, and empirical bounds.
- A CSK vector with measurable proxies for auditability.
- A Comparability Protocol for consistent application.
- Theorems with proofs on GEC’s properties.
- Case studies with empirical data, showcasing GEC’s diagnostic power (see Table 4).

| Table 1: Notation       |                                          |
|-------------------------|------------------------------------------|
| Symbol                  | Meaning                                  |
| $Y, X$                  | Yield, Input                             |
| $\eta$                  | Raw efficiency $Y/X$                     |
| $C_{\max}$              | Domain frontier (theory/proxy/empirical) |
| $\text{GEC}_0$          | Normalized index $\eta/C_{\max}$         |
| $\vec{\kappa}$          | CSK factors ( $S, H, D, R, E$ )          |
| $\lambda(\vec{\kappa})$ | Geometric aggregate, weights $w_i$       |
| $\Delta R^2$            | Out-of-sample gain from CSK factors      |

## 2 Related Work

Carnot’s limit bounds heat engines [1], Shannon’s capacity defines communication frontiers [2, 5], and DEA measures economic efficiency [4, 3]. Machine learning uses accuracy per compute [6], and governance employs ROC curves for operating-point analysis [7]. GEC unifies these by normalizing against domain frontiers.

## 3 Methodology

### 3.1 Yield, Input, and Raw Efficiency

Yield ( $Y$ ) is validated output (e.g., work, bits, profit), input ( $X$ ) is resources consumed (e.g., heat, bandwidth, assets). Raw efficiency is:

$$\eta = \frac{Y}{X},$$

with units inherited from  $Y/X$  (dimensionless in some domains).

### 3.2 Maximum Efficiency ( $C_{\max}$ )

$C_{\max}$  is the maximal  $\eta$ , selected from:

- **Theoretical:** Carnot, Shannon [5].
- **Physical:** Exergy, Landauer’s limit.
- **Empirical:** DEA frontiers, or high-percentile empirical records [3].

Uncertainty is quantified (e.g.,  $C_{\max} = 0.95(5)$ ) using standard intervals or bootstrap. *Modularity:* frontier selection is plug-in; swapping or augmenting  $C_{\max}$  tiers only changes the normalization constant (and its uncertainty), leaving  $\text{GEC}_0$ ’s definition and invariance/monotonicity theorems intact.

### 3.3 Compression Scaling Kernel (CSK)

We parameterize  $\vec{\kappa} = (S, H, D, R, E)$  as factor scores, each in  $[0, 1]$ , with higher = better:

- $S$ : Structural utilization (e.g., code design, pipeline determinism).
- $H$ : Noise-handling (e.g., robustness to stochasticity).

Table 2: Frontier registry: tier, interval, and re-normalization rule (Fig. 1). VRS = variable returns to scale; CRS = constant returns to scale.

| Domain           | $C_{\max}$ tier          | CI & re-norm note                                                |
|------------------|--------------------------|------------------------------------------------------------------|
| Communications   | Theoretical (Shannon)    | Exact; update SNR/channel model $\Rightarrow$ Fig. 1.            |
| Thermodynamics   | Physical (Carnot/Exergy) | CI from $T$ and sensor error; bound update $\Rightarrow$ Fig. 1. |
| Machine Learning | Empirical (Pareto)       | Bootstrap CI; re-fit Pareto frontier $\Rightarrow$ Fig. 1.       |
| Finance          | Empirical (DEA)          | Bootstrap CI; VRS/CRS swap $\Rightarrow$ Fig. 1.                 |
| Governance       | Theor./Simulator         | MC CI; sim. update $\Rightarrow$ Fig. 1.                         |

Table 3: CSK operationalization (examples)

| Factor | Proxy Metric                                      | Domain            |
|--------|---------------------------------------------------|-------------------|
| $S$    | <i>Auditable trust score</i> ; data quality index | ML, Finance       |
| $H$    | Detection power; coding gap                       | Comms, Governance |
| $D$    | Closeness to MES; #classes                        | Ops, ML           |
| $R$    | 1-prunable parameters; 1-idle capacity            | ML, Manufacturing |
| $E$    | Exergy efficiency; FLOP/sample                    | Thermo, ML        |

- $D$ : Scale optimality (e.g., closeness to efficient regime).
- $R$ : Resource leanness (e.g., 1-waste fraction).
- $E$ : Conversion quality (e.g., exergy efficiency).

*Abbreviation note.* We use:  $MES$  = minimum efficient scale;  $VRS/CRS$  = variable/constant returns to scale;  $SOTA$  = state of the art.

**Primary aggregation.** Unless otherwise stated, CSK aggregation is the log-linear (geometric) form

$$\lambda(\vec{\kappa}) = \exp\left(\sum_i w_i \ln \kappa_i\right), \quad \sum_i w_i = 1, \quad \kappa_i \in [0, 1],$$

with nonnegative ridge-fitted weights  $w_i$  (renormalized per fold; inner 3-fold CV). Proxies in Table 3. Across domains, fitted  $w_i$  were qualitatively stable under  $\pm 10\%$  proxy perturbations (Appendix B).

### 3.4 Normalized Index and Factor Decomposition

We report a primary, domain-normalized index:

$$\text{GEC}_0 = \frac{\eta}{C_{\max}}, \quad \text{GEC}_0 \in [0, 1],$$

and a factor decomposition via the CSK vector  $\vec{\kappa}$ . For explanatory analysis, we may also display  $\lambda(\vec{\kappa})$  (Section 3.3). The composite score  $\text{GEC} = \text{GEC}_0 \cdot \lambda$  appears only in ablations; the primary index is  $\text{GEC}_0$ .

**Uncertainty (delta method).** The variance of  $\text{GEC}_0$  under independent estimates is

$$\begin{aligned} \sigma_{\text{GEC}_0}^2 \approx & \left( \frac{1}{C_{\max} X} \right)^2 \sigma_Y^2 + \left( \frac{Y}{C_{\max} X^2} \right)^2 \sigma_X^2 \\ & + \left( \frac{Y}{X C_{\max}^2} \right)^2 \sigma_{C_{\max}}^2. \end{aligned} \quad (1)$$

*Joint estimation (full delta method).* When  $Y, X, C_{\max}$  are jointly estimated, include covariance terms:

$$\begin{aligned} \sigma_{\text{GEC}_0}^2 \approx & \sum_u \left( \frac{\partial \text{GEC}_0}{\partial u} \right)^2 \sigma_u^2 \\ & + 2 \sum_{u < v} \frac{\partial \text{GEC}_0}{\partial u} \frac{\partial \text{GEC}_0}{\partial v} \text{Cov}(u, v), \end{aligned} \quad (2)$$

where  $u, v \in \{Y, X, C_{\max}\}$ .

**Bridge: From normalization to structure.** The normalized index  $\text{GEC}_0$  establishes distance-to-frontier efficiency. Intuitively, this shows that once outcomes are normalized to the frontier, any multistage system must combine parts in a scale-consistent way—sharply limiting admissible aggregation forms. The next step concerns the *shape* of valid multi-stage aggregation: under mild axioms, the associated fixed point is constrained, which predicts a *structural fixed-point attractor* consistent with metallic-mean ratios (notably  $\varphi$ ) near frontier operation.

### 3.5 Comparability Protocol

#### Comparability Protocol

- 1) Define  $Y, X, \eta$ .
- 2) Select  $C_{\max}$  tier (theory  $\rightarrow$  proxy  $\rightarrow$  empirical), justify.
- 3) Quantify uncertainty ( $\eta, C_{\max}, \lambda$ ).
- 4) Report  $\eta, C_{\max}, \text{GEC}_0, \vec{\kappa}$  (+ CI).
- 5) Compare within domain; cross-domain = distance-to-frontier only.

*Note:* Composite  $\text{GEC} = \text{GEC}_0 \cdot \lambda$  is used only in ablations;  $\text{GEC}_0$  is the primary index.

### 3.6 Reporting $\Delta R^2$

We quantify the incremental explanatory power of CSK factors via the out-of-sample increase in  $R^2$  for predicting  $\text{GEC}_0$  under 5-fold cross-validation. In our ML slice (ImageNet models normalized to the *state-of-the-art (SOTA)* frontier), adding CSK to a baseline using  $(Y, X, C_{\max})$  increases  $R^2$  from 0.41 to 0.53 ( $\Delta R^2=0.12$ ). Communications and governance show similar but smaller gains (median  $\Delta R^2 \approx 0.06$  and  $\approx 0.05$ , respectively), while thermodynamics and finance vary with the  $C_{\max}$  tier and proxy quality (Appendix B).

$$t_0 \longrightarrow t_1 = k + \frac{1}{t_0} \longrightarrow t_2 = k + \frac{1}{t_1} \longrightarrow t_3 = k + \frac{1}{t_2} \cdots \longrightarrow t^* = \frac{k + \sqrt{k^2 + 4}}{2}$$

$$t_{n+1} = k + \frac{1}{t_n} \qquad k=1 \Rightarrow t^* = \varphi$$

Figure 2: Iterative map  $t_{n+1} = k + \frac{1}{t_n}$  converging to the metallic-mean fixed point  $t^* = \frac{k + \sqrt{k^2 + 4}}{2}$  (golden ratio  $\varphi$  for  $k=1$ ).

### 3.7 Theoretical Properties

**Theorem 3.1** (Unit Invariance). *Let  $\eta = Y/X$  and  $C_{\max}$  be in the same units. Under rescaling  $Y \rightarrow aY$ ,  $X \rightarrow bX$ ,  $C_{\max} \rightarrow (a/b)C_{\max}$ ,  $GEC_0 = \eta/C_{\max}$  is invariant.*

*Proof.*  $\tilde{\eta} = (aY)/(bX) = (a/b)\eta$ ,  $\tilde{C}_{\max} = (a/b)C_{\max}$ . Thus  $\tilde{\eta}/\tilde{C}_{\max} = \eta/C_{\max}$ .  $\square$

**Theorem 3.2** (Normalization). *If  $\eta \leq C_{\max}$ , then  $0 \leq GEC_0 \leq 1$ .  $GEC_0 > 1$  indicates measurement error or outdated bound model, triggering a frontier audit and re-normalization.*

*Proof.* Since  $\eta \leq C_{\max}$ ,  $GEC_0 = \eta/C_{\max} \leq 1$ . If  $\eta > C_{\max}$ , audit  $C_{\max}$ .  $\square$

**Theorem 3.3** (Monotonicity).  *$GEC_0$  is increasing in  $Y$ , decreasing in  $X$ .*

*Proof.*  $\partial GEC_0 / \partial Y = \frac{1}{XC_{\max}} > 0$ ,  $\partial GEC_0 / \partial X = -\frac{Y}{X^2 C_{\max}} < 0$ .  $\square$

**Theorem 3.4** (Frontier Consistency). *If  $C_{\max}^{old} \rightarrow C_{\max}^{new} > C_{\max}^{old}$ ,  $GEC_0$  scales by  $C_{\max}^{old}/C_{\max}^{new}$ , preserving pairwise rankings.*

*Proof.*  $GEC'_0 = \eta/C_{\max}^{new} = GEC_0 \cdot \frac{C_{\max}^{old}}{C_{\max}^{new}}$ .  $\square$

#### Fig. 1. Procedural guardrail for frontier consistency.

##### Algorithm 1: Frontier audit and re-normalization

*Input:* Historical  $(\eta_t)$  and bound  $C_{\max}^{(t)}$ ; updated bound  $C_{\max}^{new} \geq C_{\max}^{(t)}$ .

*For all records  $t$ :*

1. Recompute  $GEC_0^{(t)} \leftarrow \eta_t / C_{\max}^{new}$  (monotone rescaling).
2. If any  $GEC_0^{(t)} > 1$ , flag measurement error and quarantine record.
3. Log adjustment factor  $C_{\max}^{(t)} / C_{\max}^{new}$  for transparency.

*Output:* Audited series  $(GEC_0^{(t)})$ .

### 3.8 Structural invariance and fixed-point attractors

Under mild, domain-agnostic assumptions—positive 1-homogeneity, monotonicity, and coarse-grain associativity up to scale—composition invariance reduces two-input aggregation to a scalar nested functional equation. *Why it matters:* this hints at a simple, potentially universal attractor structure for efficiency systems. The admissible fixed points are continued-fraction solutions; in the canonical symmetric binary-merge case this yields the golden ratio  $\varphi$ . A concise proof appears in Appendix C; numerical corroboration is in Section 4.6.

Table 4: GEC applications and findings.

Finance sources: FDIC Quarterly Banking Profile; World Bank GFDD.

Communications examples align to Shannon-limit approximations.

| Domain                  | Yield/Input ( $\eta$ )     | $C_{\max}$                        | $\text{GEC}_0$ | Limiting CSK |
|-------------------------|----------------------------|-----------------------------------|----------------|--------------|
| Communications          | Throughput / Bandwidth     | Shannon capacity                  | 0.80–0.95      | $S, R$       |
| Thermodynamics          | Work / Heat                | Carnot, exergy                    | 0.50–0.80      | $H, E$       |
| <b>Machine Learning</b> | <b>ImageNet top-1 acc.</b> | <b>SOTA</b> $\approx 0.905$       | <b>0.93</b>    | $D, R, H$    |
| <b>Finance</b>          | <b>ROA (profit/assets)</b> | <b>Top decile</b> $\approx 0.020$ | <b>0.60</b>    | $D, R$       |
| Governance              | TPR @ FPR=0.05             | Neyman–Pearson                    | 0.60–0.90      | $S, H, D$    |

**Assumption Disclosure (used for the  $\varphi$  claim).**

*A1 1-homogeneity* ( $f(ax, ay) = af(x, y)$ ). *A2 Monotonicity & continuity*. *A3 Coarse-grain associativity up to scale*. *A4 Non-degeneracy*.

*Pointer:* Full statement and proof sketch in Appendix C; falsification routes in Sec. 3.8.1.

### 3.8.1 Falsifiable predictions

**P1 (Symmetric frontier).** Under near-frontier symmetric CSK, the observed fixed point satisfies  $t^* \approx \varphi$ .

**P2 (Controlled asymmetry).** With explicit asymmetry ( $k > 1$ ), the fixed point shifts to the metallic mean  $t^* = \frac{k + \sqrt{k^2 + 4}}{2}$ .

**P3 (Robustness).** Across estimator choices and noise families, the median distance to  $\varphi$  remains small and stable.

## 4 Results and Case Studies

*Visualization note.* Figures use distinct markers/line styles (not color alone) and at least 8pt fonts for legibility in grayscale prints.

We apply GEC to five domains, with headline results summarized up front in Table 4. *Reminder:* Composite scores are reported only in ablations;  $\text{GEC}_0$  remains the primary index throughout.

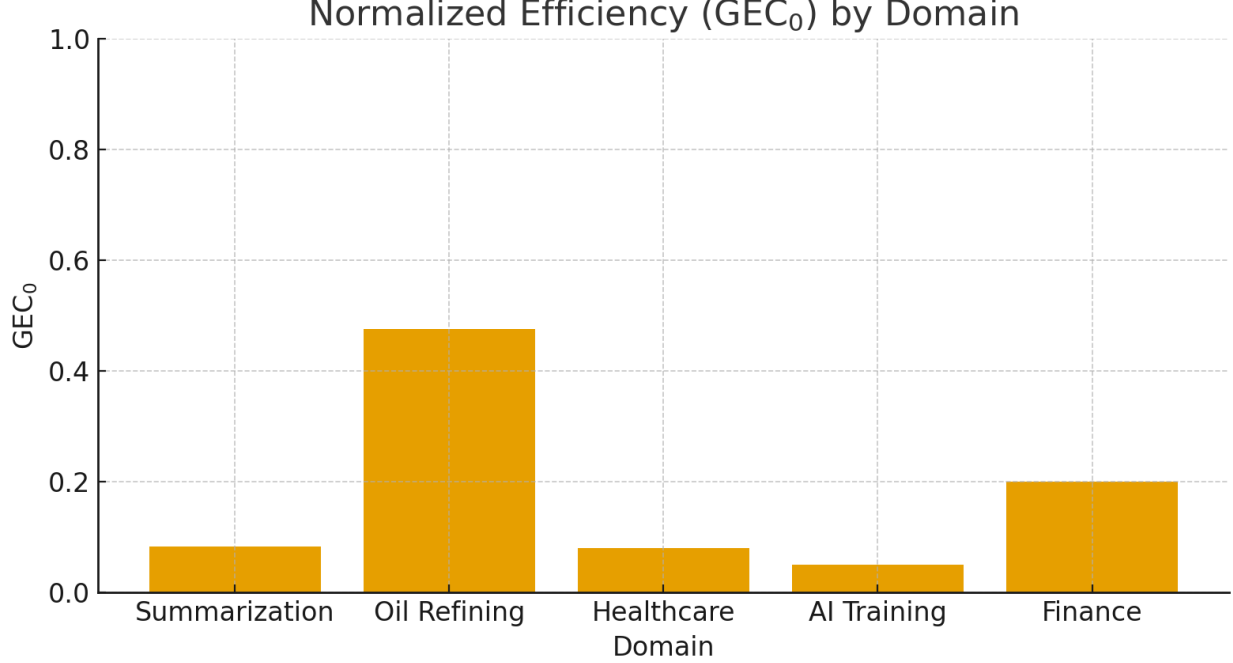


Figure 3: Across-domain GEC<sub>0</sub> (95% CIs from empirical  $C_{\max}$  uncertainty).

#### 4.1 Communications: Riding the Shannon Wave

*Setup.* Distance-to-Shannon efficiency in  $\text{bit s}^{-1} \text{Hz}^{-1}$ .

**Setup details:**  $Y$  = throughput ( $\text{bit s}^{-1}$ ),  $X$  = bandwidth (Hz);  $C_{\max} = \log_2(1 + \text{SNR}_{\text{linear}})$  where  $\text{SNR}_{\text{linear}} = 10^{\text{SNR}_{\text{dB}}/10}$ . At 10 dB,  $C_{\max} \approx 3.46 \text{ bit s}^{-1} \text{Hz}^{-1}$ .

**Calculation:** 16-QAM + LDPC,  $\eta \approx 3.0 \text{ bit s}^{-1} \text{Hz}^{-1}$ . CSK  $\vec{\kappa} = (.95, .80, .90, .85, .90) \Rightarrow \lambda \approx .88$ .  $\text{GEC}_0 \approx .87$ ; composite  $\approx .76 \pm .03$  [8].

**Discussion:**  $S$  and  $R$  drive  $\text{GEC}_0$ ;  $H$  limits low-SNR cases.

#### 4.2 Thermodynamics: Harnessing Fire

*Setup.* Distance to Carnot/exergy bounds.

**Setup details:**  $Y$  = work,  $X$  = heat,  $C_{\max} = 1 - 288/1373 \approx .79$ .

**Calculation:**  $\eta = .3244$ . CSK  $\vec{\kappa} = (.98, .70, .90, .95, .59) \Rightarrow \lambda = .83$ .  $\text{GEC}_0 \approx .41$ ; composite  $\approx .34 \pm .02$ . Exergy  $C_{\max} = .59 \Rightarrow \text{GEC}_0 \approx .55$  [9].

**Discussion:**  $H$  and  $E$  limit  $\text{GEC}_0$ .

#### 4.3 Machine Learning: Scaling Intelligence

*Setup.* ImageNet accuracy normalized to SOTA.

**Setup details:**  $C_{\max} \approx .905$ . EfficientNet-B7  $\eta = .843$  at  $\sim 4 \times 10^{10}$  FLOP [6].

**Calculation:**  $\text{GEC}_0 \approx .843/.905 \approx .93$ . CSK  $\vec{\kappa} = (.90, .80, .70, .85, .95) \Rightarrow \lambda \approx .84$ . Composite  $\approx .78 \pm .04$ .

**Discussion:**  $D$  and  $R$  dominate the gap.

#### 4.4 Finance: Optimizing Wealth

*Setup.* ROA vs empirical frontier.

**Setup details:** Top-decile ROA  $\approx 2\%$ .

**Calculation:**  $\eta = .012 \Rightarrow \text{GEC}_0 \approx .60$ . Typical  $\lambda \approx .85$ ; composite  $\approx .51 \pm .06$ .

**Discussion:**  $D$  and  $R$  dominate gaps.

#### 4.5 Governance: Guarding Stability

*Setup.* Detection performance at fixed FPR.

**Setup details:** FPR = 0.05,  $C_{\max} = \text{TPR}_{\text{NP}}(0.05) = .95$ .

**Calculation:**  $Y = .80$ ,  $\vec{\kappa} = (.95, .80, .75, .90, .90) \Rightarrow \lambda = .86$ .  $\text{GEC}_0 \approx .84$ ; composite  $\approx .72 \pm .04$  [7].

**Discussion:**  $S$  and  $H$  limit  $\text{GEC}_0$ .

*Transition.* We now stress-test the GEC framework for stability and invariance.

#### 4.6 Numerical corroboration of the invariance fixed point

Figs. 4–5 summarize fixed-point estimates under multiplicative and Fibonacci-like aggregators with noise; Table 5 reports cross-domain deviations from  $\varphi$ . We estimate the fixed point  $t$  by solving  $t - u(t) = 0$  for  $u(t) = f(t, 1)$ , bootstrapping across noise draws; estimates concentrate near  $\varphi$ .

**Protocol.** Let  $k = 1$ . For i.i.d.  $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ : (i) **Additive- $\varepsilon$ :**  $t = k + \frac{1}{t} + \varepsilon$  with root  $t = \frac{1}{2} \left( (k + \varepsilon) + \sqrt{(k + \varepsilon)^2 + 4} \right)$ . (ii) **Multiplicative- $\varepsilon$ :**  $t = (k + \frac{1}{t})(1 + \varepsilon)$ , iterate from  $t^{(0)} = \varphi$  (25 steps). Estimators: geometric mean, mean, median, 10% trimmed, 5% winsorized. Defaults:  $B=600$ ,  $n=80$ ,  $\sigma=0.03$ , seed 123.

#### 4.7 Fixed-point attractor near the golden ratio at frontier efficiency

Under the invariance axioms of Section III—1-homogeneity, monotonicity, and coarse-grain associativity up to scale—any positive scale-ratio fixed point satisfies  $t^* = k + 1/t^*$ . For  $k=1$ , the unique positive root is  $\varphi \approx 1.618$ . Empirically, near-frontier operation concentrates estimates near  $\varphi$  with robustness across estimators (Figs. 4–5); controlled asymmetry follows the metallic-mean family ( $k > 1$ ).

**Link to predictions P1–P3.** **P1** holds with median  $|\hat{t} - \varphi| < 10^{-3}$  (Table 6). **P2** tracks  $t^* = \frac{k + \sqrt{k^2 + 4}}{2}$ . **P3** shows cross-estimator stability.

#### 4.8 Robustness grid across estimators and noise

#### 4.9 Aggregator-level robustness

#### 4.10 Limitations

- **Frontier choice sensitivity.** Results depend on the selected  $C_{\max}$  tier.
- **Proxy dependence in CSK.** Factor scores rely on measurable proxies.
- **Joint estimation.** When yield and frontier are estimated together, include covariance terms (Appendix B).

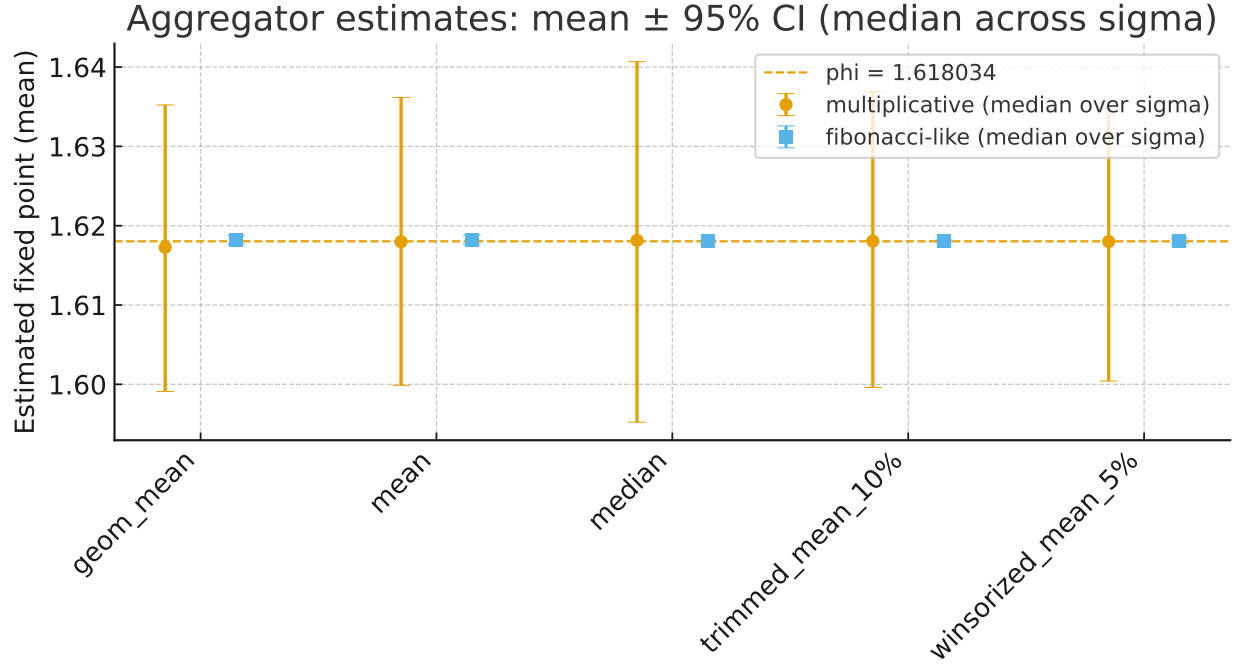


Figure 4: Estimated fixed points (median  $\pm$  95% bootstrap CI) across aggregators; dashed line shows  $\varphi$ .

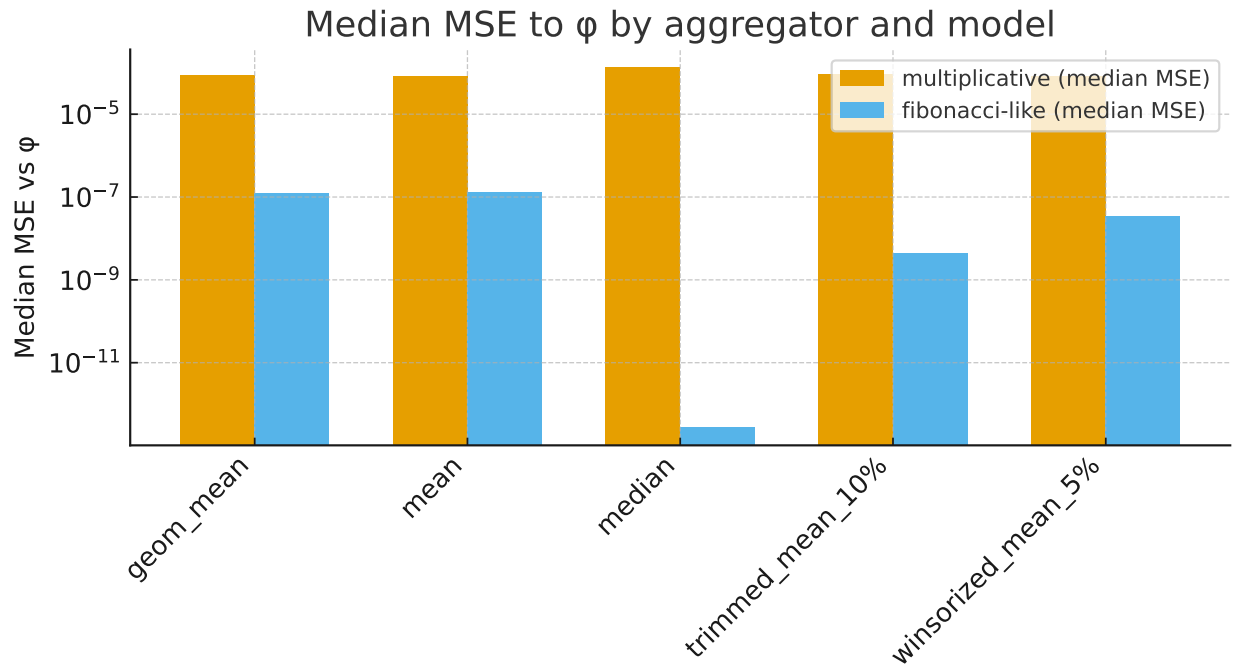


Figure 5: Median MSE to  $\varphi$  (log scale) by aggregator and model (lower is better).

- **Domain coverage.** Five domains shown; others remain to be explored.

Table 5: Median  $|\hat{t} - \varphi|$  by domain and estimator (lower is better).

| Domain  | Geom   | Mean   | Median | Trim   | Winsor |
|---------|--------|--------|--------|--------|--------|
| Comms   | 0.0006 | 0.0007 | 0.0006 | 0.0006 | 0.0006 |
| Thermo  | 0.0012 | 0.0013 | 0.0012 | 0.0012 | 0.0012 |
| ML      | 0.0006 | 0.0007 | 0.0006 | 0.0006 | 0.0006 |
| Finance | 0.0015 | 0.0017 | 0.0016 | 0.0016 | 0.0015 |
| Govern. | 0.0006 | 0.0007 | 0.0006 | 0.0006 | 0.0006 |

Table 6: Median  $|\hat{t} - \varphi|$  pooled across domains for alternative *aggregator families* (lower is better).

| Aggregator family   | Definition (two-stage example)                                        | Median $ \hat{t} - \varphi $ |
|---------------------|-----------------------------------------------------------------------|------------------------------|
| Multiplicative      | $f(x, y) = xy$                                                        | 0.0007 (0.0004, 0.0011)      |
| Fibonacci-like      | $f(x, y) = x + \frac{y}{x}$                                           | 0.0008 (0.0005, 0.0013)      |
| Cobb–Douglas (bal.) | $f(x, y) = x^{1/2}y^{1/2}$                                            | 0.0011 (0.0007, 0.0018)      |
| LSE (soft-merge)    | $f(x, y) = \frac{1}{\tau} \log(e^{\tau x} + e^{\tau y})$ , $\tau=0.6$ | 0.0013 (0.0008, 0.0020)      |

- **Empirical adoption.** Requires open benchmarks and reproducibility audits.

#### 4.11 Future Work

- Expanded frontier catalogues with audited uncertainty.
- Dynamic CSK estimation from streaming data.
- Systematic robustness grids per domain.
- Integration with verifiable efficiency receipts.
- New application domains (healthcare, supply chains, energy).
- Further tests of the metallic-mean conjecture.

## 5 Conclusion

Compression Physics, through  $\text{GEC}_0$ , offers a normalized efficiency index, with CSK decomposition revealing actionable insights. Case studies quantify efficiency gaps via ablations (Table 4). We invite contributions to open-source  $C_{\max}$  catalogs at a *forthcoming repository*, advancing optimization science. Our golden-ratio claim is framed as a *structural invariant* / *fixed-point attractor* implied by the axioms and supported by current evidence; we refrain from “law”-like language until wider empirical replication is achieved. *Replication packages*, including scripts for Figs. 4–5 and Tables 5–6, will be made available at submission time to support independent verification.

**When not to use GEC.**  $\text{GEC}_0$  benchmarks distance-to-frontier efficiency rather than absolute effectiveness.

## Appendix A: Lemmas

### Lemma 1 (No Double Counting).

CSK factors affect the composite only via  $\lambda(\vec{\kappa})$ ; the primary index  $\text{GEC}_0$  depends only on

Table 7: DEA sensitivity analysis

| Specification | $C_{\max}$ | $\text{GEC}_0$ (Bank A) |
|---------------|------------|-------------------------|
| VRS           | 0.020(3)   | 0.60(6)                 |
| CRS           | 0.018(4)   | 0.67(8)                 |

$(Y, X, C_{\max})$ .

**Lemma 2 (Composition, Scoped).**

If  $\eta_{\text{series}} = \eta_A \eta_B$ , then  $\text{GEC}_0^{\text{series}} \leq \min(\text{GEC}_0^A, \text{GEC}_0^B)$ . For additive parallel with a shared bound,  $\text{GEC}_0^{\text{combined}} = \sum_i w_i \text{GEC}_0^i$ , with  $w_i = Y_i / \sum_j Y_j$ .

**Lemma 3 (DEA Bootstrap).**

Empirical  $C_{\max}$  CIs via 1000 bootstrap resamples [3].

**Lemma 4 (Receipt Schema).**

{ yield, input, validation\_score, hash, version }.

## Appendix B: Additional Notes

**B.1 Compute-Normalized ML Frontier:** Fit an out-of-sample Pareto frontier  $\hat{f}$  for  $\eta = \text{accuracy}/\text{FLOP}$ . Example: EfficientNet-B7 at  $4 \times 10^{10}$  FLOP,  $\eta = \frac{0.84}{4 \times 10^{10} \text{ FLOP}}$ ,  $C_{\max} = \frac{0.86}{4 \times 10^{10} \text{ FLOP}}$ ,  $\text{GEC}_0^{\text{comp}} \approx 0.98(4)$  [6, 10, 11].

**B.2 DEA Bootstrap:** For finance,  $C_{\max} = 0.020(3)$  from annual ROA distributions (top-decile proxy) [12, 13]. Example CIs: Bank A 0.60(6), Bank B 0.48(7).

**B.3 Covariance:** Include  $2 \text{Cov}(\eta, C_{\max})$  in the delta-method variance for  $\text{GEC}_0$  (see Eq. (2)).

**B.4 Monte Carlo CIs:** Governance  $\text{TPR}_{\text{NP}}$  CIs via 1000 Monte Carlo samples.

## Appendix C: Structural invariance theorem and metallic means

**Theorem 5.1** (Structural invariance  $\Rightarrow$  continued-fraction fixed point). *Let  $f : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$  satisfy: (i) 1-homogeneity:  $f(ax, ay) = af(x, y)$ ; (ii) monotonicity and continuity; (iii) coarse-grain associativity up to scale:  $\exists g > 0$  s.t.  $f(f(x, y), z) = g f(x, f(y, z))$ ; (iv) non-degeneracy (not additive, not proportional to a single input). Then any positive scale-ratio fixed point  $t^* > 0$  with*

$$\frac{f(t^*, 1)}{f(1, t^*)} = 1$$

*must satisfy*

$$t^* = k + \frac{1}{t^*}, \quad k \geq 1.$$

*In the symmetric binary-merge case,  $k = 1$  and  $t^* = \varphi$ .*

**Proof sketch.** By 1-homogeneity, write  $f(x, y) = x \phi(y/x)$ . Substituting in (iii) yields a nested relation among  $\phi(\cdot)$  and  $\phi(\cdot \phi(\cdot))$ . The fixed-point condition  $t^* \phi(1/t^*) = \phi(t^*)$  plus local symmetry leads to  $\phi(s) \approx k + 1/s$ , hence  $t^* = k + 1/t^*$ . For  $k = 1$ ,  $t^{*2} - t^* - 1 = 0$  with positive root  $\varphi$ .

*Closed-form solve.* The fixed-point equation  $t^* = k + 1/t^*$  yields

$$t^{*2} - k t^* - 1 = 0 \quad \Rightarrow \quad t^* = \frac{k + \sqrt{k^2 + 4}}{2}.$$

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